NON-SMOOTHABILITY AND MIRROR NON-ISOLATEDNESS OF BATYREV CALABI–YAU THREEFOLDS

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Abstract. Let $\Delta \subset \mathbb{Z}^4$ be a reflexive polytope with unit edges, so that the generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ is a Calabi–Yau threefold whose singular points are the cones over the two-dimensional faces of Δ. We ask when the Batyrev mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ is *also* isolated-singular, and relate this to the local models of the prequels. We first record an exact identity: the number $\ell(F^\circ)$ of singular points of X over a two-dimensional face F equals the multiplicity of the transverse cone of F; consequently both X and X° are isolated-singular — we call Δ *both-sides unit* — if and only if every two-dimensional face of Δ is transversally smooth. We then argue that the novel non-smoothable germs of the prequels — the anticanonical cone over $\mathbb{F}_1$, and the deformable-but-non-smoothable pentagon cone — cannot occur on a both-sides unit polytope. Precisely, we conjecture that *every two-dimensional face of a both-sides unit reflexive* 4*-polytope is a triangle or a centrally-symmetric polygon*, and we support this by a complete check of the 3,905,789 reflexive 4-polytopes with at most nine vertices and of the non-smoothable-face consequence for the 3,735,806 polytopes with at least 23 vertices, by the explicit structure of the both-sides examples — including the 36-vertex hexagon-product polytope at the top of the classification, which is both-sides unit, conforms, and mirrors a smooth Calabi–Yau threefold — and by a reduction of the conjecture to a purely local lemma about a face and its star. The conjecture implies that the Batyrev mirror of a Calabi–Yau threefold carrying a non-quotient or a deformable non-smoothable point is always non-isolated, so that its singularities leave the isolated Altmann–Corti–Filip–Petracci framework and are governed instead by Filip's non-isolated 0-mutability criterion. All computations are performed by the exact-arithmetic programs of the prequels, extended by a scanner included as ancillary files.

1. Introduction

Let $\Delta \subset N_\mathbb{R} = \mathbb{R}^4$ be a reflexive polytope all of whose edges have lattice length one, let $\Delta^\circ \subset M_\mathbb{R}$ be its polar dual, and let $X \in |-K_{\mathbb{P}_\Delta}|$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ. By Batyrev [2] X is a Calabi–Yau threefold with Gorenstein canonical singularities, and by the planting proposition of [6] its singular locus is exactly: for each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, $\ell(F^\circ)$ points at which the germ of X is analytically isomorphic to $C(F)$, where $\ell(F^\circ)$ is the lattice length of the edge F° of Δ° dual to F. The trichotomy of [6] classifies $C(F)$ by the combinatorics of F into rigid (R), deformable-but-non-smoothable (D), and smoothable (S); the first two are non-smoothable, and a single such face forces X to admit no smoothing.

Mirror symmetry, in the Batyrev picture, is the polar duality $\Delta \leftrightarrow \Delta^\circ$, which for 4-polytopes exchanges two-dimensional faces with edges. The mirror family is the generic anticanonical $X^\circ \subset \mathbb{P}_{\Delta^\circ}$, whose singular locus is governed in the same way by the two-dimensional faces of Δ°. The purpose of this note is to understand what the non-smoothability of X — that is, the presence of a face of type (R) or (D) — means for X°, and in particular when X and X° are *both* isolated-singular.

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Our first observation is exact and elementary. For a two-dimensional face F write $u_1, u_2 \in M$ for the primitive inner normals of the two facets of Δ containing F, so $F^\circ = \operatorname{conv}\{u_1, u_2\}$, and let $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$, a rank-two saturated sublattice.

Theorem 1.1 (= Lemma 3.1)**.** *$\ell(F^\circ) = [\,M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2\,]$, the multiplicity of the transverse cone* $\operatorname{Cone}(u_1, u_2)$ *of F. In particular $\ell(F^\circ) = 1$ if and only if u_1, u_2 form a basis of M_F, i.e. the transverse cone is smooth.*

Since X is isolated-singular exactly when Δ has unit edges, and X° exactly when Δ° does — equivalently when every two-dimensional face F of Δ has $\ell(F^\circ) = 1$ — Theorem 1.1 gives a clean reading of the doubly-isolated condition.

Corollary 1.2. *X and X° are both isolated-singular if and only if Δ has unit edges and every two-dimensional face of Δ is transversally smooth. We call such Δ* both-sides unit*.*

The point-packets of X and the singular curves of X° correspond exactly: the $\ell(F^\circ)$ points of type $C(F)$ over F are mirror to a single curve of transverse $A_{\ell(F^\circ)-1}$ singularities on X° along the toric curve dual to F (§4). Thus X° is isolated precisely when no such curve appears, i.e. on a both-sides unit polytope.

Our main assertion is that the non-smoothable local models introduced in the prequels are incompatible with this doubly-isolated situation, except in the classical quotient case. Call a lattice polygon a *zonotope* if it is a Minkowski sum of lattice segments; a unit-edge polygon is a zonotope if and only if it is centrally symmetric, and every such polygon is smoothable of type (S).

Conjecture 1.3 (triangle-or-zonotope)**.** *Every two-dimensional face of a both-sides unit reflexive* 4*-polytope is either a triangle or a zonotope.*

Equivalently: on a both-sides unit polytope every non-smoothable two-dimensional face is a triangle, hence a cyclic quotient singularity. We are unable to prove Conjecture 1.3, but we give four kinds of evidence (§5): a complete verification over the 3,905,789 reflexive 4-polytopes with at most nine vertices (exactly five both-sides unit polytopes, all conforming); a verification of its non-smoothable-face consequence over the 3,735,806 polytopes with at least 23 vertices, where the sweep of the prequels recorded no both-sides polytope carrying a non-smoothable face at all; the explicit mirror-pair structure of the both-sides examples — including a new pair far beyond the low-vertex range, the 36-vertex product of two reflexive hexagons (the largest reflexive 4-polytope by vertex count) and its 12-vertex polar, whose faces are zonotopes and smooth triangles respectively; and a reduction of the conjecture to a purely local statement about a face and its star, verified on its 38,571 candidate instances over the polytopes with at most eight vertices. We also explain why the natural stronger guess — that a both-sides unit polytope is simplicial — is *false*, by a self-duality argument (Remark 5.1).

The geometric consequence is the following dichotomy, which we regard as the main message of the note; it is unconditional on the quotient side and conditional on Conjecture 1.3 on the other.

Corollary 1.4 (mirror non-isolatedness)**.** *Let Δ be reflexive with unit edges and let X have a non-smoothable singular point whose germ is* not *a cyclic quotient singularity — for instance the anticanonical cone over $\mathbb{F}_1$ of* [6, Thm. 5.1]*, or the deformable-but-non-smoothable pentagon cone of* [6, Thm. 5.3]*. Then, assuming Conjecture 1.3, the Batyrev mirror X° is non-isolated. Consequently the mirror singularities of the novel non-smoothable Calabi–Yau* *threefolds of* [6] *always lie outside the isolated Altmann–Corti–Filip–Petracci framework and are governed by Filip's non-isolated* 0*-mutability criterion* [4].

A clean two-sided, isolated-to-isolated correspondence between non-smoothable Calabi–Yau threefolds therefore exists *only* in the classical quotient (triangle) stratum, and even there the both-sides examples pair a singular threefold with a smooth one rather than singular with singular (Example 5.2). The natural home of the mirror question for the new phenomena is thus the non-isolated theory of [4], and, conjecturally, the mirror-theoretic language of 0-mutable Laurent polynomials [3] in which non-smoothability should read as a combinatorial obstruction (§6).

All computations in this note — the transverse multiplicities, the both-sides scans of the Kreuzer–Skarke list (a reference scanner and a fast-engine scanner whose rare hits are re-verified by the reference toolkit), the 36-vertex mirror pair, and the local reduction — are implemented in the exact-arithmetic programs of [6], extended by the scanners described in §5; they are included with the arXiv submission as ancillary files.

Acknowledgements. This note continues [6, 7]; the question of smoothing canonical Calabi–Yau singularities was posed to the author by Mark Gross many years ago.

2. Preliminaries

We keep the notation of [6]. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold; Q is of type (R), (D) or (S) according as its edge-vector multiset has no proper zero-sum sub-multiset, has one but no partition into unit segments and unimodular triples, or has such a partition [1, 3]. For a reflexive polytope Δ we write Δ° for the polar dual; a face $G \preceq \Delta$ has a dual face $G^\circ \preceq \Delta^\circ$ with $\dim G + \dim G^\circ = 3$, so two-dimensional faces F of Δ are dual to edges F° of Δ° and vice versa. We write $\ell(\cdot)$ for lattice length. By the planting proposition [6, Prop. 3.1], for Δ with unit edges the generic $X \in |-K|$ meets the toric curve $V(\sigma_F)$ of a two-dimensional face F in $\ell(F^\circ)$ reduced points, at each of which the germ of X is $C(F)$; X is smooth away from these points, so X is isolated-singular exactly when Δ has unit edges.

3. The transverse-multiplicity identity

Let F be a two-dimensional face of a reflexive polytope $\Delta \subset N_{\mathbb{R}}$, and let $u_1, u_2 \in M$ be the primitive inner normals of the two facets of Δ containing F, so that $\langle u_i, \cdot\rangle = -1$ on F and $F^\circ = \operatorname{conv}\{u_1, u_2\}$. Put $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$, a rank-two saturated sublattice of M containing u_1, u_2 as primitive vectors.

Lemma 3.1. $\ell(F^\circ) = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$.

Proof. Both sides depend only on the pair (u_1, u_2) inside the rank-two lattice M_F. Choose a $\mathbb{Z}$-basis of M_F in which $u_1 = (0, -1)$; this is possible because u_1 is primitive. Write $u_2 = (p, q)$; then $p = \det(u_1, u_2) = \pm[\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$, and $\ell(F^\circ) = \ell\big(\operatorname{conv}\{u_1, u_2\}\big) = \gcd(u_2 - u_1) = \gcd(p,\, q + 1)$.

Let v be a vertex of F and let $\bar v \in N_F := N/(N \cap u_1^\perp \cap u_2^\perp)$ be its image; since u_1, u_2 vanish on $N \cap u_1^\perp \cap u_2^\perp$ they descend to the dual basis of $M_F = N_F^\vee$, and $\bar v$ is a lattice point of $N_F \cong \mathbb{Z}^2$ with $\langle u_i, \bar v\rangle = -1$. Writing $\bar v = (a, b)$ in the dual basis, $\langle u_1, \bar v\rangle = -b = -1$ gives $b = 1$, and $\langle u_2, \bar v\rangle = pa + q = -1$ gives $pa = -1 - q$, so $p \mid (q + 1)$. Hence $\gcd(p, q + 1) = |p|$, and $\ell(F^\circ) = |p| = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$. $\square$

Remark 3.2. Lemma 3.1 is elementary and may be known; what matters here is the reading it provides. The right-hand side is the multiplicity of the two-dimensional cone $\mathrm{Cone}(u_1, u_2)$, which is the normal cone of F and dually the transverse cone of Δ along F (the image of the tangent cone of Δ at an interior point of F under the quotient $N_{\mathbb{R}} \to N_{\mathbb{R}}/\operatorname{lin}(F)$). Thus *the dual-edge length counts exactly the transverse singularity multiplicity*, and $\ell(F^\circ) = 1$ is the statement that Δ is transversally smooth along F. The lattice-point hypothesis on v is essential: for an arbitrary pair of primitive vectors u_1, u_2 the two sides of Lemma 3.1 differ.

Corollary 1.2 is immediate: X is isolated-singular iff Δ has unit edges (§2); X° is isolated-singular iff Δ° has unit edges, i.e. every edge F° of Δ° has $\ell(F^\circ) = 1$, i.e. by Lemma 3.1 every two-dimensional face of Δ is transversally smooth.

4. The mirror correspondence of the singular strata

We record the mirror-dual description of the singular loci, which puts Corollary 1.2 in geometric terms. It is a direct consequence of Batyrev's construction and the planting proposition, and was the starting point of this investigation.

Proposition 4.1. *Let Δ be reflexive with unit edges and F a two-dimensional face with $C(F)$ singular. Under polar duality F corresponds to the edge F° of Δ°, and:*

1. *X has $\ell(F^\circ)$ isolated points of type $C(F)$ over F;*
2. *X° has, along the toric curve $V(\sigma_{F^\circ})$ dual to F°, a curve of transverse $A_{\ell(F^\circ)-1}$ singularities (and none if $\ell(F^\circ) = 1$).*

In particular X° is isolated-singular if and only if $\max_F \ell(F^\circ) = 1$, i.e. Δ is both-sides unit.

Proof. (1) is [6, Prop. 3.1]. For (2), the edge F° of Δ° has lattice length $\ell(F^\circ)$, so it carries $\ell(F^\circ)-1$ interior lattice points; the corresponding toric curve $V(\sigma_{F^\circ}) \subset \mathbb{P}_{\Delta^\circ}$ meets the generic anticanonical X° in a 1-dimensional stratum along which $\mathbb{P}_{\Delta^\circ}$ has a transverse $A_{\ell(F^\circ)-1}$ singularity (an edge of lattice length m produces a curve of transverse A_{m-1} singularities, [6, §2]). Hence X° is smooth in codimension two along $V(\sigma_{F^\circ})$ precisely when $\ell(F^\circ) = 1$, and it is isolated-singular iff this holds for every two-dimensional face F, i.e. iff Δ is both-sides unit. ☐

Thus the point-packet of X over F and the $A_{\ell(F^\circ)-1}$-curve of X° are mirror-dual, and the two threefolds are simultaneously isolated-singular exactly on the both-sides unit polytopes.

5. Both-sides unit polytopes and the triangle-or-zonotope conjecture

We now study which local models can occur on a both-sides unit polytope. Recall that a unit-edge polygon is a *zonotope* — a Minkowski sum of unit segments — if and only if it is centrally symmetric (its edge vectors occur in antipodal pairs), and that every zonotope is of type (S): the antipodal pairs partition the edge multiset into unit segments, giving a smoothing. Thus Conjecture 1.3 asserts that the only non-smoothable faces of a both-sides unit polytope are triangles.

Remark 5.1 (why not "simplicial"). It is tempting to conjecture the stronger statement that every two-dimensional face of a both-sides unit polytope is a *triangle*, i.e. that Δ is simplicial. This is false. The both-sides unit condition is self-dual: it holds for Δ if and only if it holds for Δ°. Were every two-dimensional face of a both-sides unit polytope a triangle, applying this to both Δ and Δ° would make Δ simplicial and simple at once, hence a simplex — contradicting the existence of both-sides unit polytopes with more than five vertices (there is one with nine, Example 5.2). The non-triangular faces of a both-sides unit polytope are genuine; the content of Conjecture 1.3 is that they are *smoothable* (centrally symmetric), never of the new non-smoothable types.

Example 5.2 (the both-sides examples are mirror pairs)**.** Among the 3,905,789 reflexive 4-polytopes with at most nine vertices, exactly five are both-sides unit; they are closed under polar duality (two polar pairs and one self-polar polytope). One pair is the smooth simplex and the simplex $\Delta = \mathrm{conv}\{(1,0,0,0),(0,1,0,0),(2,4,5,0),(3,3,0,5),(-6,-8,-5,-5)\}$, whose ten two-dimensional faces are all rigid triangles (cyclic quotient points) and whose polar is smooth. A second pair consists of a simplicial six-vertex polytope, all of whose eighteen faces are smooth triangles, and its polar (nine vertices), whose fifteen faces are six rigid triangles and nine centrally-symmetric quadrilaterals of type (S) with two interior points. In every case the two-dimensional faces are triangles or zonotopes, and every non-smoothable member is paired under the mirror with a *smooth* threefold, not a singular one. These are exactly the classical quotient examples one dimension beyond which [6] was written.

Example 5.3 (a both-sides pair at the top of the classification)**.** Both-sides unit polytopes are not confined to few vertices. The 36-vertex product $H \times H$ of two reflexive hexagons — the reflexive 4-polytope with the most vertices — is both-sides unit, and its polar, the 12-vertex free sum $H^\circ \oplus H^\circ$, is the mirror partner guaranteed by self-duality of the both-sides condition. The faces of $H \times H$ are 36 unit parallelograms and 12 centrally-symmetric dP_6 hexagons — all zonotopes, so Conjecture 1.3 holds — and the corresponding X is singular with 48 isolated points, all of type (S) (36 ordinary double points and 12 dP_6-cone points). The faces of the polar are 72 smooth triangles, so the mirror X° is a *smooth* Calabi–Yau threefold. The pattern of Example 5.2 repeats at the opposite end of the classification: a singular member (here everywhere locally smoothable) mirrors a smooth one. All assertions are machine-checked in `missing_polytope.py`.

5.1. **Computational evidence.** The exact-arithmetic classifier of [6], applied to a two-dimensional face together with its two facet normals, computes $\ell(F^\circ)$ and the type of F; a scan over the Kreuzer–Skarke list [5] then tests Conjecture 1.3 directly. Over all 3,905,789 reflexive 4-polytopes with at most nine vertices we find no counterexample: exactly five polytopes are both-sides unit, and every two-dimensional face of each is a triangle or a centrally-symmetric zonotope. (Both scanners — the reference one and a fast-engine one capable of the higher vertex counts — are among the ancillary files; every both-sides hit of the fast scanner is re-verified by the reference toolkit.)

Three further pieces of data come from the full classification. First, the complete sweep of [6] tracked, for every polytope with at least 23 vertices (3,735,806 of them), whether it is both-sides unit *and* carries a non-smoothable face: none is. So a counterexample to Conjecture 1.3 with a *non-smoothable* asymmetric face — the only kind relevant to Corollary 1.4 — has, in addition, no instance with $v \geq 23$. Second, the unique reflexive 4-polytope with more than 33 vertices — the 36-vertex product $H \times H$ of two reflexive hexagons, the one polytope absent from the per-vertex-count mirror used in [6] — is both-sides unit, and conforms: its 48 two-dimensional faces are 36 unit parallelograms and 12 dP_6 hexagons, all centrally symmetric (ancillary file `missing_polytope.py`). Third, its polar — the 12-vertex free sum $H^\circ \oplus H^\circ$ — is then also both-sides unit, with all 72 two-dimensional faces smooth triangles; see Example 5.3.

5.2. **Reduction to a local lemma.** The obstruction underlying Conjecture 1.3 is local to a face and its star. Say a two-dimensional face F is *asymmetric* if it is neither a triangle nor a zonotope, and call the two-dimensional faces sharing at least one vertex with F the *star* of F.

Conjecture 5.4 (local balancing)**.** *Let F be an asymmetric two-dimensional face of a reflexive 4-polytope such that F and every face in its star have unit edges, and $\ell(F^\circ) = 1$. Then some face in the star of F has $\ell(\cdot^\circ) \geq 2$.*

Conjecture 5.4 implies Conjecture 1.3: on a both-sides unit polytope all edges are unit and all dual-edge lengths equal 1, so no asymmetric face can occur. It is strictly local — it constrains only F and its star, ignoring the rest of Δ — and correspondingly it is a stronger statement, with far more instances. Over the reflexive 4-polytopes with at most eight vertices (1,037,834 polytopes) there are 38,571 asymmetric faces with a unit-edge star and $\ell(F^\circ) = 1$, and Conjecture 5.4 holds for every one of them: each has a star face with $\ell(\cdot^\circ) \geq 2$.

Two structural facts point toward a proof. First (Lemma 3.1), for a face F with an interior lattice point m, every facet of Δ through an edge e of F other than the two facets through F has inner-normal (x_1, x_2)-part a positive multiple of the outer edge-normal $e^\perp$: the facets adjacent to F across its edges lean strictly inward. Second, comparing a zonotope face admitting an all-ℓ=1 star with an asymmetric face shows the former places each edge of F on the minimal number (3) of facets, while the latter forces some edge onto more, carrying the excess into a dual edge of length ≥ 2. A rigorous local proof — in particular of the quadrilateral case, where “asymmetric” means “rigid” — is the natural next step and is left open.

6. Consequences and questions

Corollary 1.4 follows from Proposition 4.1 and Conjecture 1.3: a non-quotient non-smoothable germ $C(F)$ has F non-triangular and of type (R) or (D), hence not a zonotope, so Δ is not both-sides unit and X° is non-isolated. The two flagship germs of [6] are covered: the anticanonical cone over $\mathbb{F}_1$ is the rigid quadrilateral, and the deformable-but-non-smoothable cone is the pentagon $Q = T + s$; neither is a triangle or a zonotope.

Remark 6.1. The unconditional part of the story is already definite. By Proposition 4.1 and Lemma 3.1, whenever a unit-edge Δ has *any* two-dimensional face F with $\ell(F^\circ) \geq 2$ the mirror X° is non-isolated; and the both-sides scan shows this is the generic situation — among the unit-edge reflexive 4-polytopes with at most nine vertices that carry a non-smoothable face, all but two have $\max_F \ell(F^\circ) \geq 2$, and the two exceptions are both-sides unit with only triangular (quotient) non-smoothable faces; among the polytopes with at least 23 vertices the sweep of [6] found no both-sides polytope with a non-smoothable face at all. Conjecture 1.3 upgrades these two exceptions to “the exceptions are exactly the quotient stratum.”

Question 6.2. Prove Conjecture 5.4 (hence Conjecture 1.3). The quadrilateral case asks: can a rigid unit-edge quadrilateral occur as a two-dimensional face of a reflexive 4-polytope with $\ell(F^\circ) = 1$ and a unit, all-ℓ=1 star? The evidence says no.

Question 6.3. Corti–Filip–Petracci [3] relate the smoothing components of $C(Q)$ to the 0-mutable Laurent polynomials with Newton polygon Q, a mirror-theoretic statement, and Filip [4] treats the non-isolated case. In that language, what is the mirror-theoretic meaning of a face of type (R) or (D)? Corollary 1.4 says these are precisely the faces whose mirror is non-isolated; one expects non-smoothability to appear as the absence of a 0-mutable Laurent polynomial, and it would be interesting to read the trichotomy off the canonical scattering diagram of the mirror.

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